

Skills Summary

Intervals of Convergence, Endpoint by Endpoint

Use this reference while completing your worksheet or when studying.

Skill 1 — The radius from the coefficients

Rule: for $\sum c_n(x-a)^n$, the radius is $R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$. Convergence is certain on $(a - R, a + R)$ and impossible outside it. Two special cases: $R = \infty$ means every real x works, and $R = 0$ means only $x = a$ does. Neither has endpoints to test.

Example: Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n 4^n}$.

Step 1. The coefficients are $c_n = 1/(n 4^n)$, so the radius comes straight from the limit of their ratio — no need to carry x through the ratio test.

Step 2. $\left| \frac{c_n}{c_{n+1}} \right| = \frac{(n+1)4^{n+1}}{n 4^n} = \frac{4(n+1)}{n}$, whose limit is the constant part.
 $\Rightarrow \mathbf{R = 4}$

Try it: Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n 7^n}$.

check: $R = 7$

Skill 2 — Testing an endpoint

Substitute the endpoint back into the original series and test the ordinary series you get. The ratio test is useless here — it returns 1 at every endpoint by construction. The three you will meet again and again: $\sum \frac{1}{n}$ diverges, $\sum \frac{(-1)^n}{n}$ converges, and $\sum \frac{1}{n^p}$ converges exactly when $p > 1$. The two endpoints of one series can easily disagree.

Example: For $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius 1, centre 0), test the endpoint $x = -1$.

Step 1. Substituting the endpoint turns the power series into an ordinary series, and the alternating signs point at the alternating series test.

Step 2. At $x = -1$ the series is $\sum \frac{(-1)^n}{n}$: the sizes $\frac{1}{n}$ decrease to zero, so it converges, and its value is known.

$\Rightarrow -\ln 2 \approx -0.693$, so this endpoint is included

Try it: For $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$, test the endpoint $x = 1$ and give the sum.

check: $\frac{9}{2}$, so this endpoint is included

Skill 3 — When the variable carries a coefficient

Factor the constant out of the shifted power first. A series in $(kx - b)^n$ is a series in $(x - a)^n$ with $a = b/k$ and coefficients multiplied by k^n — so its radius is k times smaller than a careless reading suggests. Then finish as usual: endpoints at $a \pm R$, each tested on its own, square bracket for a convergent endpoint and round for a divergent one.

Example: Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{(2x - 8)^n}{n}$.

Step 1. Rewrite $(2x - 8)^n$ as $2^n(x - 4)^n$ so the series is visibly centred at 4, with coefficients $c_n = 2^n/n$.

Step 2. $\left| \frac{c_n}{c_{n+1}} \right| = \frac{2^n}{n} \cdot \frac{n+1}{2^{n+1}} = \frac{n+1}{2n}$, whose limit is the reciprocal of the factored constant.
 $\Rightarrow \mathbf{R} = \frac{1}{2}$, so the interval is centred at 4 with half-width one half

Try it: Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{(4x - 12)^n}{n}$.

check: $\frac{4}{1} = \mathbf{R}$